

DecodeLabs Virtual Internship

Project 1: Data Cleaning & Preparation

Intern Name: Asma Yousaf

Objective

The objective of this project was to clean a raw dataset by identifying and handling missing values, checking duplicate records, correcting data formats, and ensuring data consistency for further analysis.

Tools Used

- Microsoft Excel

Dataset Overview

The dataset contains customer order information with the following attributes:

- OrderID
- Date
- CustomerID
- Product
- Quantity
- UnitPrice
- ShippingAddress
- PaymentMethod
- OrderStatus
- TrackingNumber
- ItemsInCart
- CouponCode
- ReferralSource

- TotalPrice

Data Cleaning Process

1. Missing Values

- Identified missing values in the CouponCode column.
- Replaced blank CouponCode values with "**No Coupon**" to improve consistency.
- No missing values were found in the remaining columns.

2. Duplicate Records

- Checked the OrderID column for duplicate records.
- No duplicate Order IDs were found.

3. Data Format Validation

- Verified that all dates followed the **YYYY-MM-DD** format.
- Verified that CustomerID and OrderID followed a consistent format.

4. Numerical Data Validation

Validated the following numerical columns:

- Quantity
- UnitPrice
- ItemsInCart
- TotalPrice

Verified that TotalPrice correctly matched the calculation:

TotalPrice = Quantity × UnitPrice

5. Text Consistency

Verified consistent values in the following columns:

- Product
- PaymentMethod
- OrderStatus
- ReferralSource

No spelling inconsistencies were found.

Conclusion

The dataset was successfully cleaned and validated. Missing values were handled appropriately, duplicate records were not found, data formats were consistent, and numerical values were verified. The dataset is now clean, accurate, and ready for further analysis.